City, University of London



School of Science & Technology

EE2600 Digital Design

Project Report

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Abstract

The project aims to design a digital system for monitoring and comparing heart rate readings from two distinct biomedical sensors: an ECG and a pulse oximeter. The primary goal is to ensure that both sensors provide consistent readings, with the system identifying any significant discrepancies that may indicate errors or failures in the sensors using Logisim. By comparing the subtracted value from the ECG and Pulse oximeter reading against a selected error threshold, if it exceeds more than 2 times, it proves the either the ECG or Pulse oximeter is faulty.

Introduction

In clinical practice, monitoring heart rate is a vital activity, with its measurement being integral for patient care. This initiative is focused on engineering a digital system to validate the functionality of two heart rate monitoring devices: the Electrocardiogram (ECG) and the Pulse Oximeter. The ECG is a well-established method for directly tracking the electrical activity of the heart, providing immediate access to heart rate data [1]. On the other hand, Pulse Oximetry is an indirect, non-invasive approach that measures the oxygen saturation level of the blood through photoplethysmography [1]. By utilizing two distinct light wavelengths, it captures the absorption variance which is influenced by the pulsatile nature of blood flow, thereby allowing the determination of heart rate [1]. This project aims to harness Logisim circuits to simulate the comparative analysis of data from both sensors, ensuring they agree and, consequently, are functioning correctly. The resultant system will not only measure heart rate but will also confirm the reliability of these measurements by cross-referencing the outputs from both the ECG and Pulse Oximeter, thus enhancing the trustworthiness of heart rate monitoring in healthcare settings.

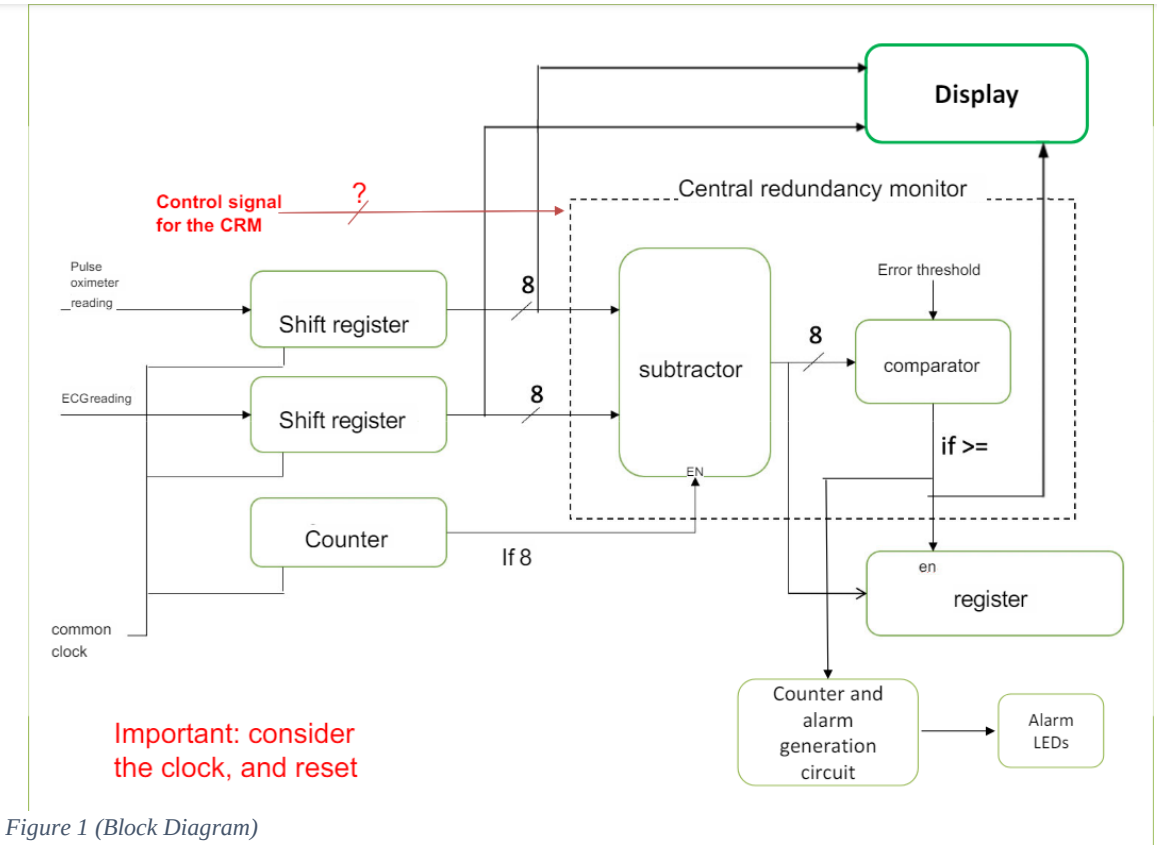


Figure 1 (Block Diagram)

Specification

Signal Input

The system's operation commences with the signal input phase, where it receives 8-bit values transmitted serially from two different types of biomedical sensors: a pulse oximeter and an ECG.

- **Pulse Oximeter Input**: Captures blood oxygen levels and heart rate.
- **ECG Input**: Records the electrical activity generated by heart contractions.

Sensor data is input serially into shift registers, synchronized by a common clock, consolidating 8-bit values for subsequent analysis, ensuring stable and synchronized processing inputs.

CRM (Signal Processing)
The central redundancy monitor (CRM) constitutes the system's signal processing core. It takes the parallel outputs from the shift registers—now holding the 8-bit values from the pulse oximeter and ECG—and performs a comparative analysis.

- **Subtractor**: A combinational logic circuit within the CRM takes the two 8-bit numbers and calculates the difference between them.
- **Comparator**: The output of the subtractor, which represents the calculated difference, is then passed to a comparator. The comparator assesses whether this difference lies within a predefined tolerance range, effectively determining the accuracy and reliability of the sensor data.

Through these operations, the CRM processes and interprets the inputs, ascertaining the consistency of the data and highlighting any potential errors. It ensures that only accurate and trustworthy data is utilized for clinical decision-making.

Alert and Display
The final phase of the system revolves around the alert and display mechanisms, which are predicated on the CRM's findings.

- **Alert System**: If discrepancies are detected and they exceed the error tolerance, the system escalates the alert levels through amber and red indicators, signaling the need for immediate attention or intervention depending on the counter.
- **Display**: Alongside the alert system, a display unit presents the status of the readings. It may show the actual sensor values, the difference between them. Value stored in 8-bit registers.

Together, the alert system and the display create a feedback loop that not only notifies but also informs healthcare providers about the data's integrity, ensuring patient safety and the proper functioning of the biomedical sensors.

Methodology

8 Bit Shift Register

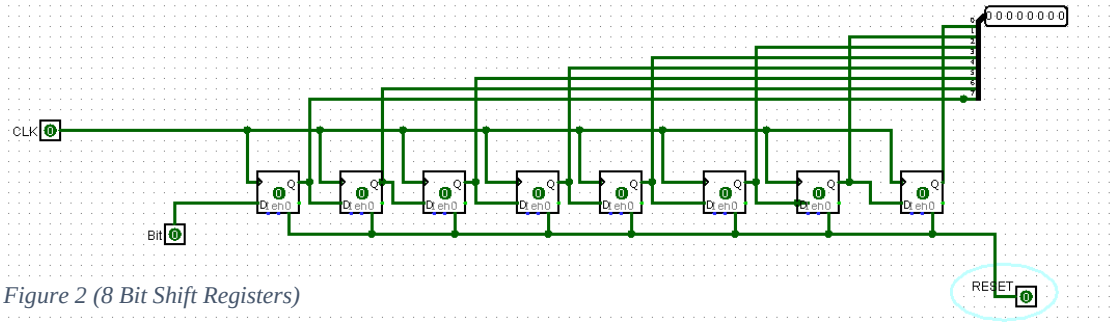


Figure 2 (8 Bit Shift Registers)

For the construction of the 8 Bit Shift Register we used 8 D flip flops, which are edge triggered on the rising edge of the clock cycle. Each D flip-flop stores one bit of data, the simplicity of D flip-flop's design having a single data input and data output [2], facilitates the straightforward data transfer ideal for serial to parallel data conversion that shift register performs. We connect the Q output of each D flip flop into the D input which allows the right shift of the data. Lastly, compared to JK or T flip flops which require multiple external control signals to manage their states, D flip-flops only require a reset, reducing complexity and potential for signal errors. The counter ensures all 8 clock cycles are counted before the data is passed onto the next stage.

Counter

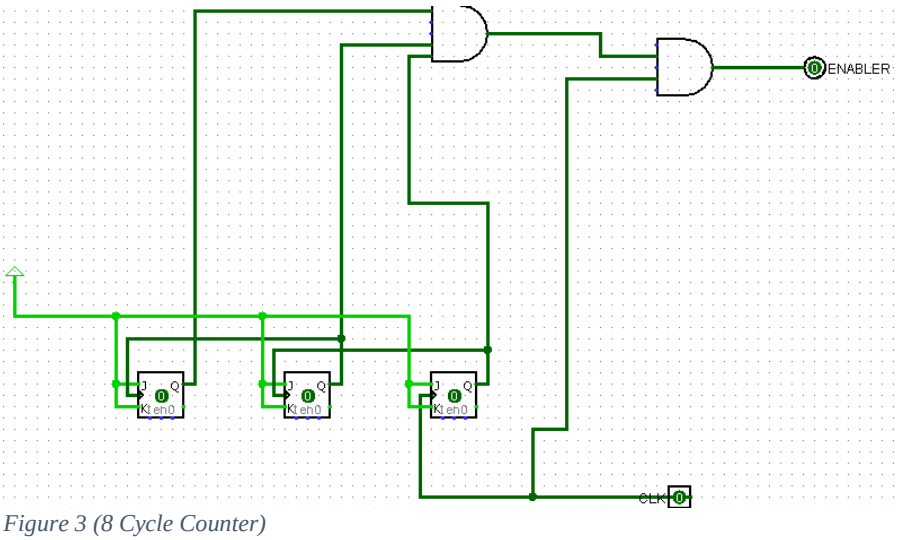


Figure 3 (8 Cycle Counter)

For the counters we have used JK flip flops due to their unique properties, their ability to toggle their output on the application of the input signal, when both J & K inputs are high, the output of the JK flip-flop changes state on every clock pulse [3]. This characteristic allows designs of counter with minimum number of external components. Furthermore, the design of a counter that resets after reaching a specific count (8), the counter can be made to automatically reset to 0 after the 8th count. The reason we have 3 JK flip flops are due to the need of having 2^3 = 8 unique states and when it reaches the state of 111 to switch on the enabler for the subtractor.

Subtractor

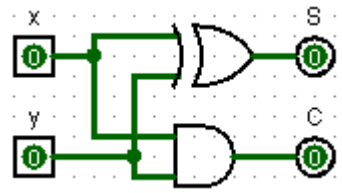


Figure 4(Half Adder Circuit)

x	y	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Figure 5(Half Adder Truth Table)

Building the 8 bit-subtractor from gate level design requires the design of a **half adder** which is used to add two single-bit binary number and produce a sum and carry bit. Choosing to use the XOR gate for the sum and an AND gate for the carry. The limitation of the half adder doesn't allow the carry from the previous addition to carry over [4].

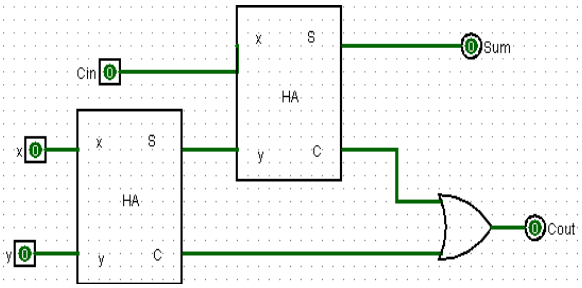


Figure 6 (Full Adder Circuit)

Cin	x	y	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Figure 7(Full Adder Truth Table)

The **full adder** extends the capabilities of the half adder. The inclusion of an additional input for carry bits, allow for multi-bit binary addition. The full adder is built by the combination of 2 half adders and an OR gate to handle the carry bits [4].

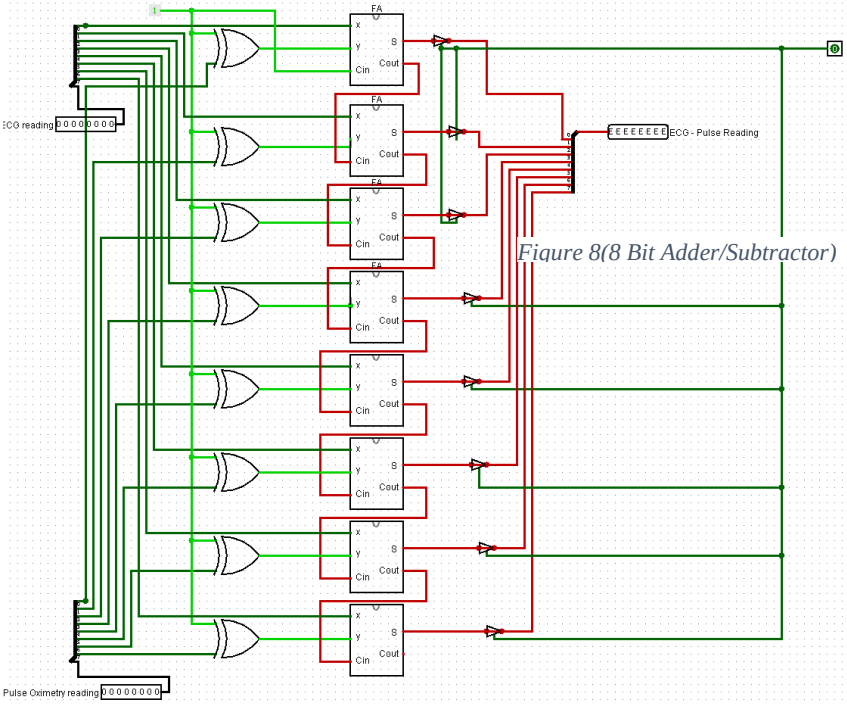


Figure 8(8 Bit Adder/Subtractor)

The subtraction of 2 binary numbers can be done by taking the 2's compliment of the subtrahend, adding to the minued [4]. The XOR gate acts as an inverter, when one input of an XOR gate is held high (the mode) which we need for subtraction. The other input is inverted, XOR gate only outputs high when 1 of the 2 inputs are unique from each other. Considering the case where B is larger value than A, this would lead to a negative 8-bit value. We've prevented this from happening by introducing a comparator and multiplexer which compares the value B in respect to A. Where appropriate ensures the larger value is the minued and the smaller value is the subtrahend. The selection line is based off the comparator in which the case is B>A would invert the input of the subtractor.

The enable signal coming from the counter ensures that the values from each of the full adder are only passed on after the full 8 clock cycles. This control is reinforced by the controlled inverter when it's HIGH the values are passed as ERRORS. Whereas when

it's LOW the values are passed through to the output. Ensuring values are only compared when it's ready [5].

Comparator

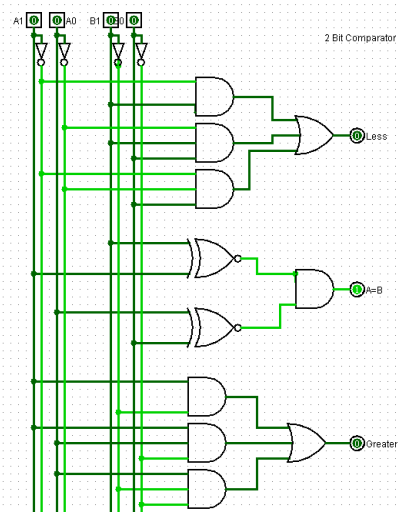


Figure 9 (2 Bit Comparator)

A1	A0	B1	B0	Less	AB	Greater
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0

Figure 10 (2 Bit Comparator Truth Table)

In essence, the 2-bit comparator takes two sets of inputs, compares corresponding bits using XOR gates for equality, and uses AND/OR gate combinations to establish the greater/less than relationships. The output from these gates is then combined to give the final comparison results [6].

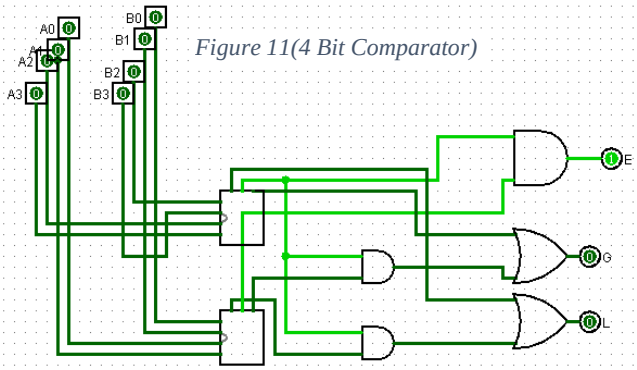
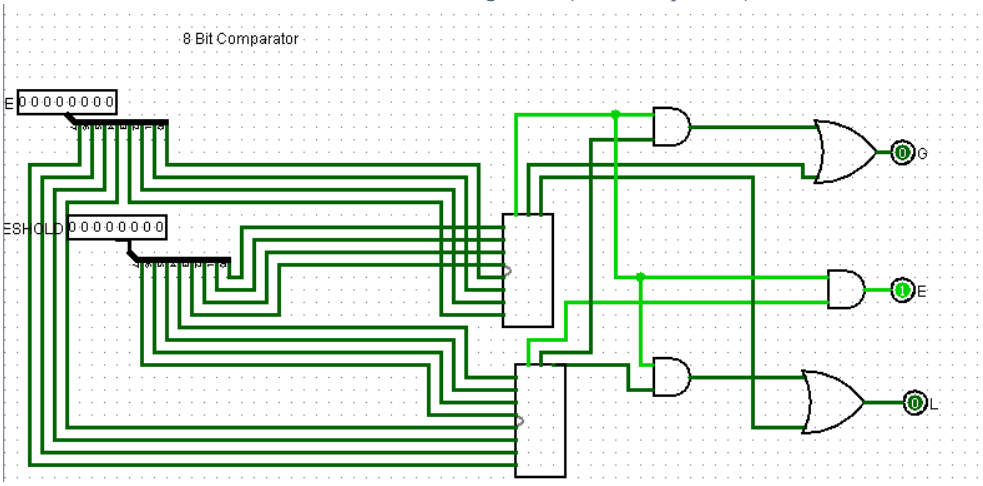


Figure 11(4 Bit Comparator)

The 4-bit comparator is created using two 2-bit comparators, the output of the 2-bit stages is used to determine the final 4-bit comparison.
In comparator circuits, 'greater than' or 'less than' signals propagate from MSB to LSB. Propagation logic would be implemented by using the gates to chain outputs from one stage to the next, ensuring that the comparison continues to the less significant bits only if the more significant bits are equal [6].

Figure 12 (8 Bit Comparator)



The 8-bit comparator built with AND and OR gates utilize a hierarchical design, wherein pairs of 4-bit comparators form the foundation. Each 4-bit comparator assesses its segment of the binary numbers, outputting 'equal', 'greater than', and 'less than' signals. The Priority logic is about establishing which part of the data inherently carries more weight in the final decision-making process. It sets the rules for which signals should be prioritized in affecting the output. unless equality prevails. In cases of equality, OR gates combine the lower-order

comparator outcomes to finalize the 8-bit comparison, thus ensuring correct prioritization [6].

PIPO 8 Bit Register

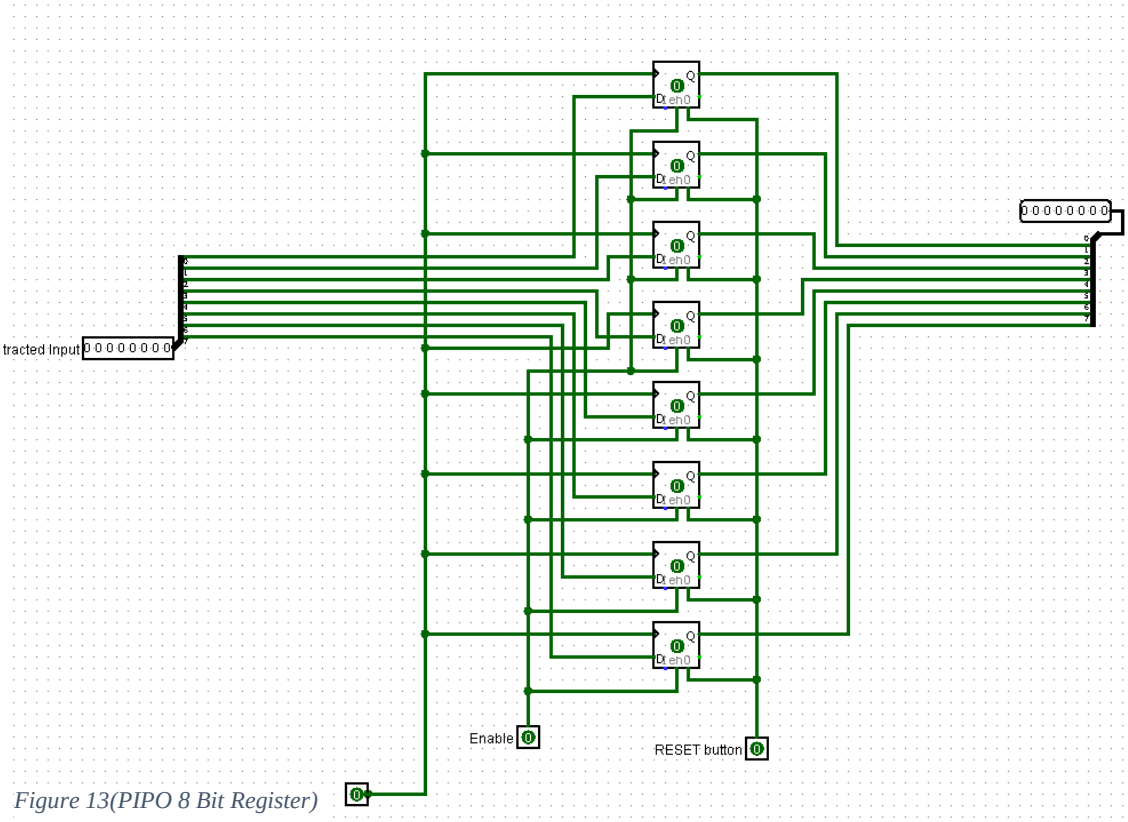


Figure 13(PIPO 8 Bit Register)

In the design of the 8-bit Parallel-In Parallel-Out (PIPO) register within the digital system, D-type flip-flops (DQ flip-flops) are employed to store and maintain binary data consistently. Each of the eight D flip-flops in the register is responsible for capturing and holding a single bit from the 8-bit input data, aligning precisely with each rising clock pulse that triggers data capture. In addition to that the enable signal, comes from the comparator when the subtracted value is larger or equals to the error threshold value which must also be on HIGH for data capture. The parallel input and output architecture of the PIPO register allows for all eight bits to be written to or read from simultaneously [7], making this register particularly efficient for operations requiring quick data retrieval or update without shift delays.

Alarm & LED System

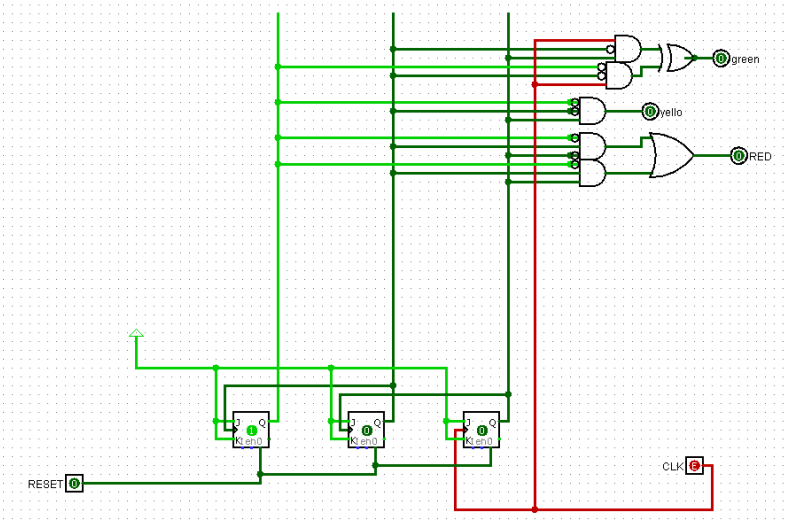


Figure 14 (Alarm generation circuit)

In the alarm system the implementation of 3 JK flip flops were sufficient to display all possible LED states. It acts as a counter with integrated LED system which selectively turns on each LED light depending on the comparator and how many high signals (1) it sends to the input. Adding additional logic to the circuit ensures that the appropriate LED is turned on/off when needed.

A2	A1	A0	Fg	Fy	Fr
0	0	0	1	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	0	0	1
1	0	0	0	0	1
1	0	1	0	0	1
1	1	0	0	0	1
1	1	1	0	0	1

Figure 15 (Truth Table For Alarm)

- 000 would display GREEN (given the comparator turns on the counter)
- 001 would display YELLOW.
- 011 Would display All 3 colours indicating 4 cycles exceeded threshold values.

Final circuit

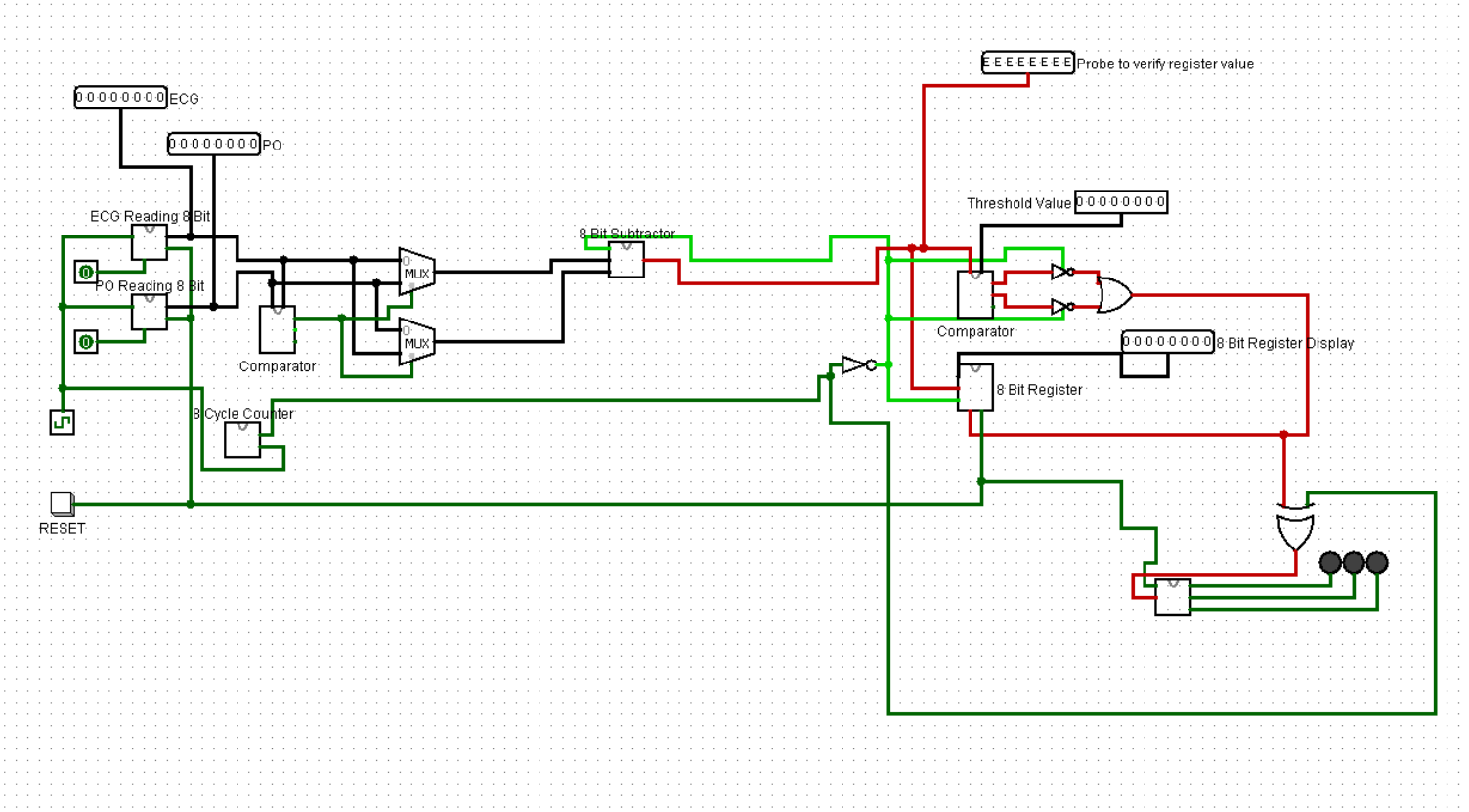


Figure 16 (Final Circuit)

Testing Input (8 Bit Shift Register FOR ECG + Comparator + Multiplexer)

Input	Clock Cycle	Expected Output	Actual Output
0	1	00000000	00000000
1	2	10000000	10000000
1	3	11000000	11000000
1	4	11100000	11100000
1	5	11110000	11110000
0	6	01111000	01111000
0	7	00111100	00111100
0	8	00011110	00011110

Using the probe, it shows that with each clock cycle the registers are acting accordingly right shifting all the data bits by 1. The expected output and actual output proving it's operating correctly. Furthermore, the comparator which acts as the control for the subtractor to pick the correct inputs for which value is being subtracted also works.

Testing CRM (Subtractor)

Input	Scenario	Expected Output	Actual Output

11111111-00000000	1	11111111	11111111
00000000-11111111	2	11111111	11111111

Testing the operation of our subtractor and whether inputs would be inverted based off the multiplexer, proves that it's able to compute and would not return negative values.

Testing CRM (Comparator)

Input	Scenario	Expected Output	Actual Output
11111111v00000000	1	GREATER	GREATER
00000000v11111111	2	LESS	LESS
11111111v11111111	3	EQUAL	EQUAL

Testing the operation of our comparator would compare the value of the subtracted input and the error threshold. It's able to correctly compute and draw a comparison between the two 8-bit value. Enabling our 8-bit register, using the probe function on the subtracted wire, and displaying the 8-bit value output for the register, they match up proving that our register correctly stores the value.

Testing the alarm generation

Input	Scenario	Expected Output	Actual Output
1st Threshold Exceeded	1	Green LED ON	Green LED ON
2nd Threshold Exceeded	2	Yellow LED ON	Yellow LED ON
3rd Threshold Exceeded	3	Red LED ON	Red LED ON
4th Threshold Exceeded	4	Red LED ON	Red LED ON

Testing our alarm generation circuit, it performs the function we need it to and meets expectations.

Conclusion

This project successfully designed a digital system to compare heart rate measurements from ECG and Pulse Oximeter sensors, addressing a critical need in clinical settings. Employing Logisim, the system integrated an 8-bit subtractor for two's complement arithmetic, and a comparator to avoid negative outcomes, all culminating in a reliable alarm system.

Challenges like error wires were encountered, signaling potential logical issues, yet these were essential in the iterative design process, enabling the identification and resolution of problems to ensure compliance with the project specifications. The system's ability to correctly perform operations despite these initial errors demonstrates the robustness of the design process. However, challenges may arise when scaling up the project to implement more advanced functions.

Use of D flip-flops to create stable registers, and multiplexers for dynamic input control, highlights the project's adherence to digital design principles. The result is a testament to the efficacy of a methodical approach in digital system design.

References

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